

Exercise 4 (CHLMOT003)

Claude AI was used
to guide and correct
and explain my mistakes.

$$1.) f(x) = \frac{x}{x+1}$$

$$2.) f(x) = \frac{2x^2}{x^4}$$

$$3.) f(x) = \ln x^2$$

$$1. f(x) = \frac{x}{x+1}$$

$$f(x_0) = \frac{x_0}{x_0+1}$$

$$f'(x) = \frac{(1)(x+1) - (x)(1)}{(x+1)^2} = \frac{1}{(x+1)^2}$$

$$f'(x_0) = \frac{1}{(x_0+1)^2}$$

$$f''(x) = -2(x+1)^{-3} = \frac{-2}{(x+1)^3}$$

$$f''(x_0) = \frac{-2}{(x_0+1)^3}$$

$$f(x) = \frac{x_0}{x_0+1} + \frac{1}{(x_0+1)^2} (x-x_0) - \frac{1}{(x_0+1)^3} (x-x_0)^2 + \mathcal{O}((x-x_0)^3)$$

$$2. f(x) = \frac{2x^2}{x^4}$$

$$f(x) = \frac{x^2 \cdot 2}{x^4}$$

$$= 2x^{-2}$$

$$f(x_0) = 2x^{-3}$$

$$= \frac{2}{x_0^3}$$

$$f'(x) = -2x^{-3}$$

$$= 2(-2)x^{-2}$$

$$= -4x^{-3} = \frac{-4}{x_0^3}$$

$$f''(x) = -4x^{-3}$$

$$= -3(-4)x^{-4}$$

$$= 12x^{-4} = \frac{12}{x_0^4}$$

$$f(x) = \frac{2}{x_0^2} - \frac{4}{x_0^3} + \frac{6}{x_0^4}$$

Claude AI assisted
with this breakdown

$$3) f(x) = \ln x^2$$

$$f'(x) = \frac{2}{x}$$

$$f'(x_0) = \frac{2}{x_0}$$

$$f''(x) = -2x^{-2}, \quad -2/x^2$$

$$f''(x_0) = \frac{-2}{(x_0)^2}$$

$$f(x) = \ln(x_0^2) + \frac{2}{x_0}(x-x_0) - \frac{1}{x_0^2}(x-x_0)^2 + \mathcal{O}((x-x_0)^3)$$